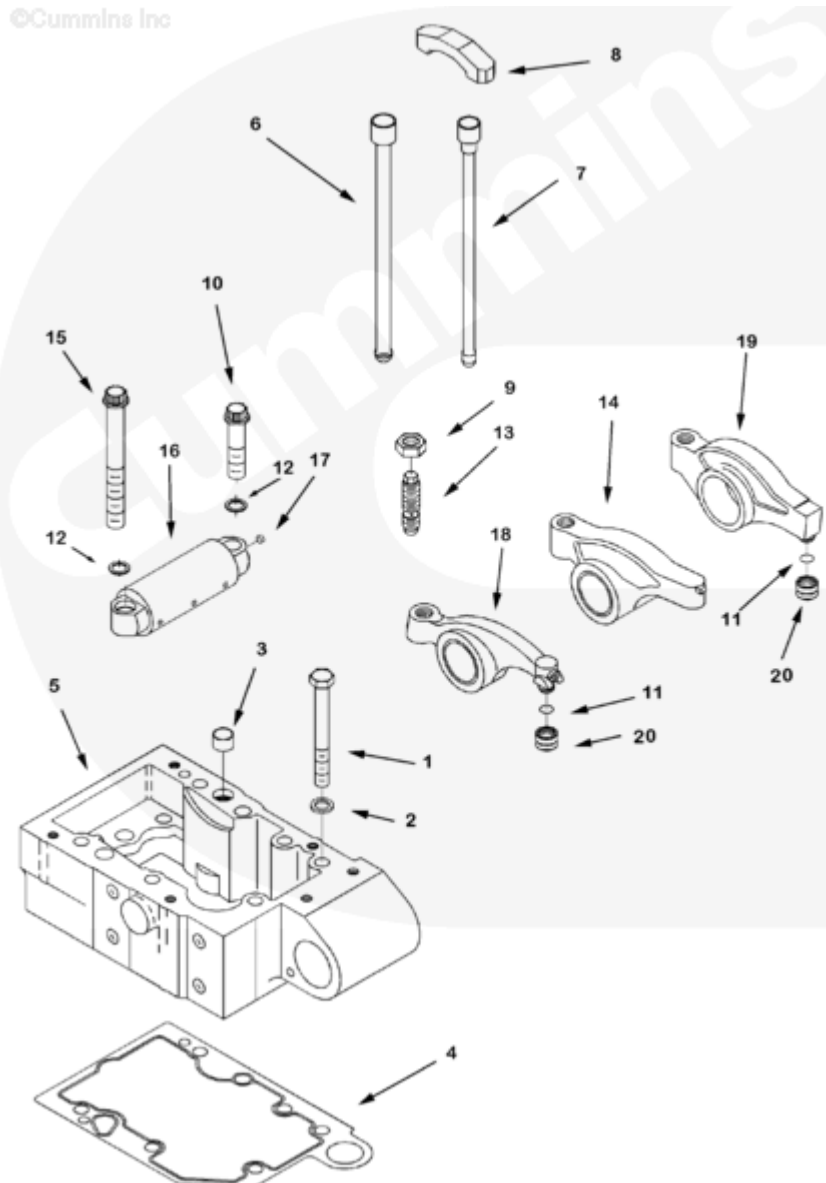


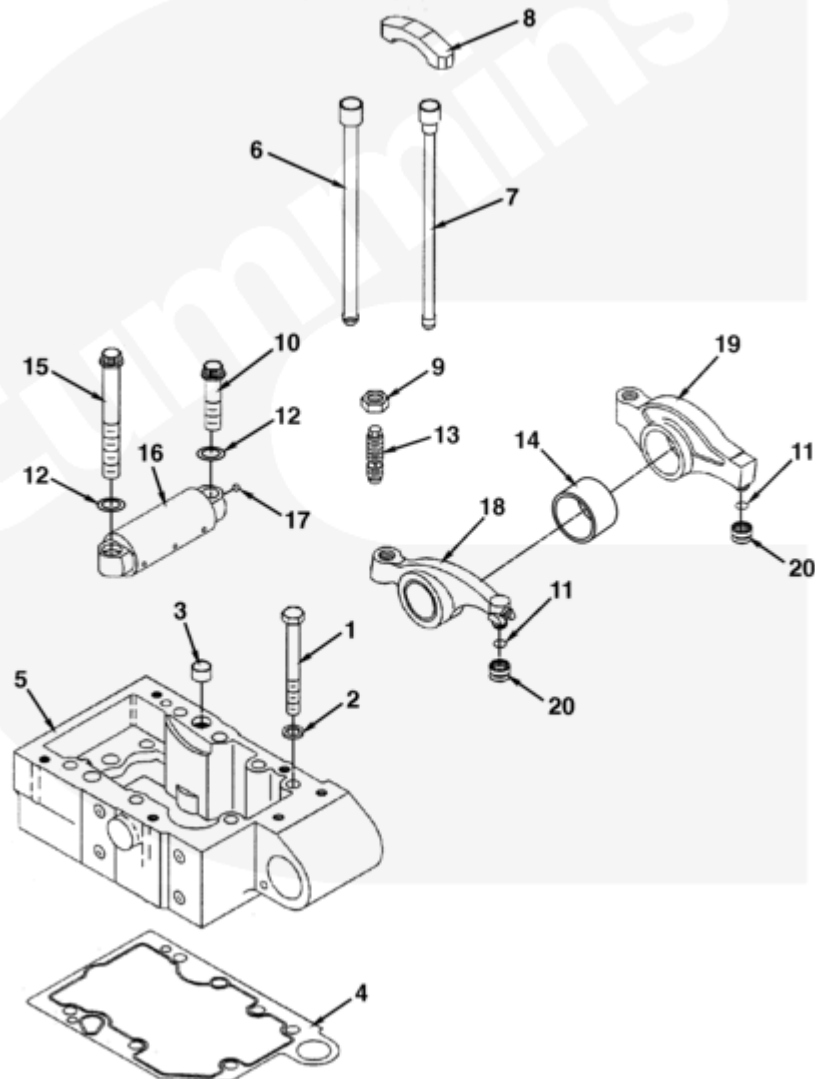
Exploded View

with Mechanically Actuated Injector



1. Hexagon head capscrew
2. Plain washer
3. Dowel ring
4. Rocker lever housing gasket
5. Rocker lever housing
6. Push rod
7. Push rod

8. Valve crosshead
9. Regular hexagon head jam nut
10. Twelve point capscrew
11. O-ring seal
12. Plain washer
13. Rocker lever adjusting screw
14. Rocker lever
15. Twelve point capscrew
16. Rocker lever shaft
17. Expansion plug
18. Rocker lever
19. Rocker lever
20. Rocker lever socket.



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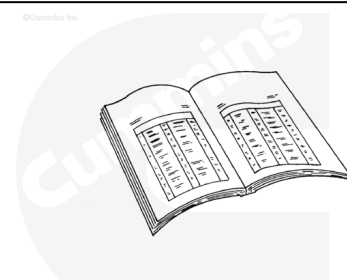
4. Rocker lever housing gasket
5. Rocker lever housing
6. Push rod
7. Push rod
8. Valve crosshead
9. Regular hexagon head jam nut
10. Twelve point capscrew
11. O-ring seal
12. Plain washer
13. Rocker lever adjusting screw
14. Spacer
15. Twelve point capscrew
16. Rocker lever shaft
17. Expansion plug
18. Rocker lever
19. Rocker lever
20. Rocker lever socket.

Preparatory Steps

Industrial and Power Generation

- Remove the rocker lever covers. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>).

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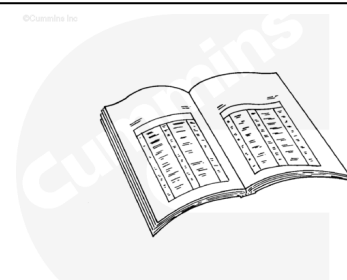


ck800wa

QSK60 Marine Applications

- Remove the aftercooler heat shields. Refer to Procedure 010-129 (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>).
- Remove the rocker lever covers. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>).

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ck800wa

with Electronically Actuated Injector

- Remove the injector supply lines. Refer to Procedure 006-051 (/qs3/pubsys2/xml/en/procedures/56/56-006-051.html).
- Remove the rocker lever covers. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).



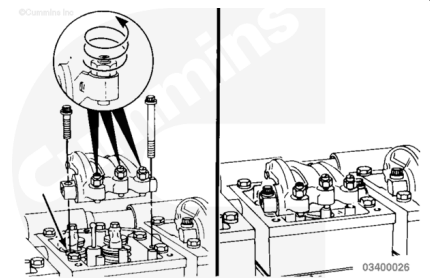
Remove

with Mechanically Actuated Injector

Do **not** allow the rocker levers to fall off the shaft during removal.

Remove the two rocker lever assembly mounting capscrews.

Remove the rocker lever assembly.

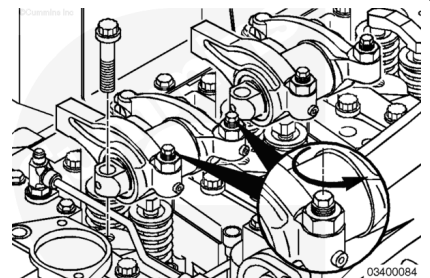


with Electronically Actuated Injector

Loosen the rocker lever adjusting screws.

Loosen the rocker lever shaft mounting capscrews and remove the rocker lever assembly.

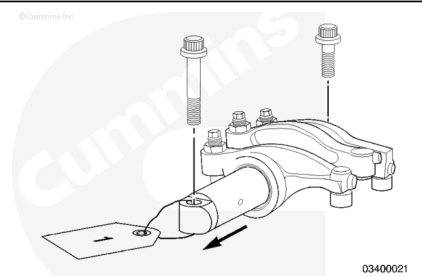
Do **not** allow the rocker levers to fall off the shaft during removal.



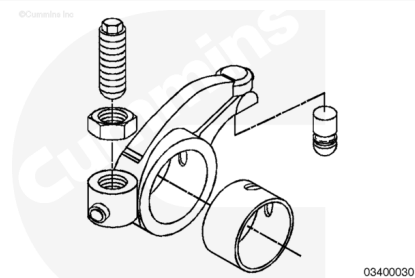
Disassemble

with Mechanically Actuated Injector

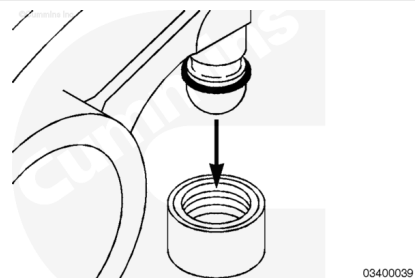
Remove the capscrews. Pull the shaft out. Tag the parts to aid in the assembly.



Remove the lock nut and adjusting screw.



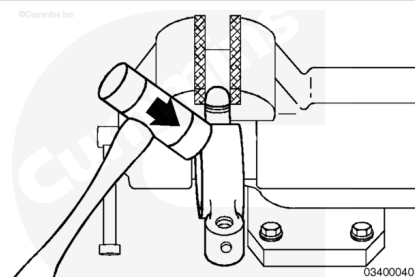
Remove the socket from the ball on the rocker lever.



Place the ball in a vise as shown.

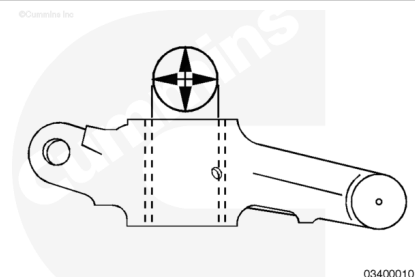
Wiggle the rocker lever until the ball swivels.

Use a mallet to tap the rocker lever until the ball comes out of the rocker arm.



Check the bushing wear by measuring the inside diameter of the bushing.

Rocker Lever Bore Inside Diameter (Bushing Installed)		
mm		in
47.66	MIN	1.876
47.73	MAX	1.879



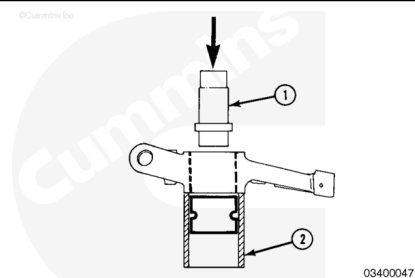
This specification applies to all rocker levers.

Do **not** remove the bushing unless it is damaged or worn.

If cracks are suspected in the rocker lever remove the bushing to perform magnetic crack inspection.

Support the lever (2) as shown. Use an arbor press and rocker lever mandrel, Part Number 3162458, (1) or equivalent.

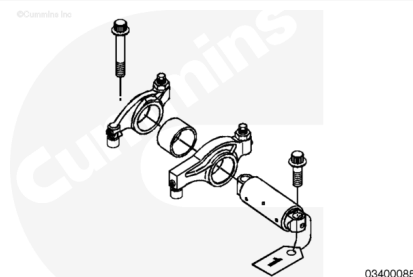
Remove the bushing.



with Electronically Actuated Injector

Remove the capscrews.

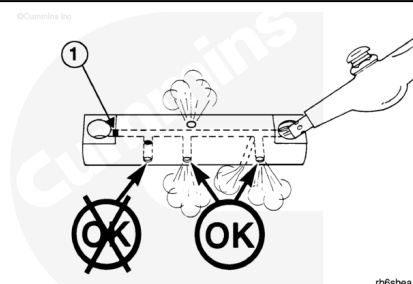
Pull the shaft out of the rocker levers.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

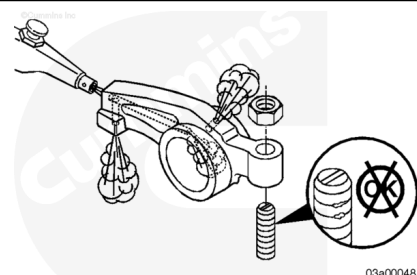
Use solvent to clean the shaft.

Use compressed air to blow through the oil drillings to be sure that the nine oil holes are **not** blocked.

The plug (1) in the main rocker lever shaft **must** be installed in the ring dowel end of the shaft (exhaust lever end).

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a 240-grit, or finer, emery cloth to remove any burrs from the bore of the rocker lever.

Clean the rocker lever support with a steam cleaner or solvent.

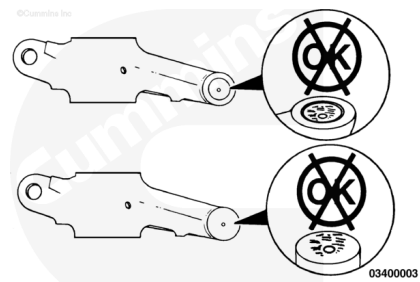
Use compressed air to blow through the oil drillings to be sure the drillings are **not** restricted.

Inspect the adjusting screw threads for damage. Replace if needed.

Inspect the socket for wear on both the injector and valve sockets.

Inspect the crosshead for wear.

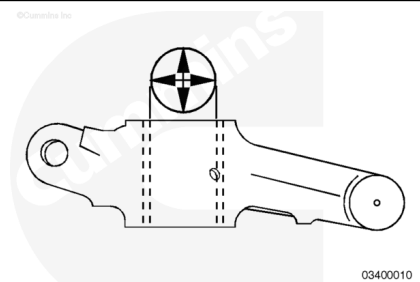
If there is damage that can be felt with the fingernail the parts **must** be replaced.



Inspect the bushing for wear.

Measure the inner diameter of the bushing.

Rocker Lever Bushing Bore Inner Diameter (Bushing Installed)		
mm		in
47.66	MIN	1.876
47.73	MAX	1.879

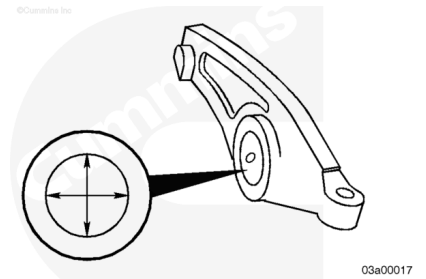


This specification applies to all rocker levers.

Measure each rocker lever bore inside diameter.

Rocker Lever Bore Inside Diameter (Bushing Removed)

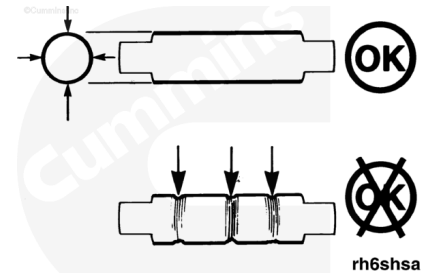
mm		in
49.246	MIN	1.939
49.270	MAX	1.940



Note : The rocker lever shaft for the mechanically actuated injector will show a wear pattern as shown. The electronically actuated injector rocker lever shafts will **only** show a wear pattern on the two outer areas of the shaft.

Inspect the shaft for damage or roughness.

Measure the outside diameter.



Rocker Arm Shaft Outside Diameter

mm		in
47.592	MIN	1.874
47.634	MAX	1.875

Magnetic Crack Inspect

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



The rocker lever **must** be disassembled to perform this inspection.

Use the residual method and apply coil shot amperage.

Ampere turns is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

Amperage (Ampere Turns)

1200 VDC or rectified VAC

2000 VDC or rectified VAC

The lever **must** be replaced if there is a crack in any location.

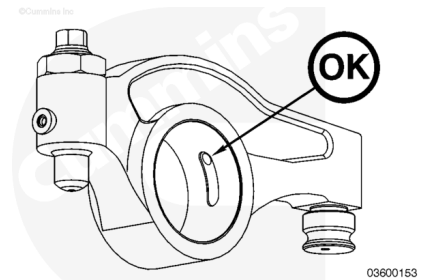
Demagnetize the lever.

Use solvent to clean the rocker lever.

Assemble

⚠ CAUTION ⚠

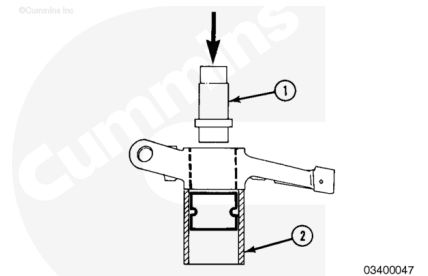
The bushing must be installed with the long slot in the direction of the crosshead side of the lever (elephant's foot side). A loss of lubricating oil to the elephant's foot and crosshead will occur if the bushings are incorrectly installed.



Align the oil holes in the bushing with the oil passages in the rocker lever.

⚠ CAUTION ⚠

The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.

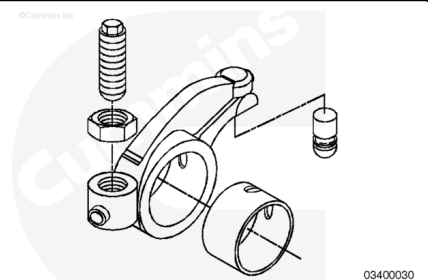


Support the rocker lever as shown (2).

Use an arbor press and a rocker lever mandrel (1), Part Number 3162458, or equivalent to install the bushing.

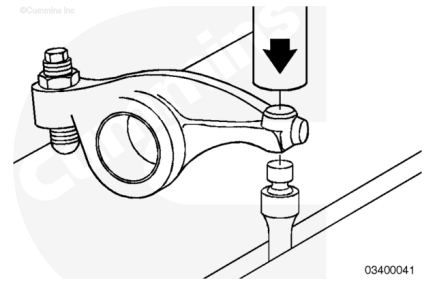
Install the adjusting screw and the locknut in the rocker lever.

Do **not** completely tighten the locknut until the assembly is installed on the engine.



⚠ CAUTION ⚠

Do not overstress the press fit. Damage to the rocker lever can result.



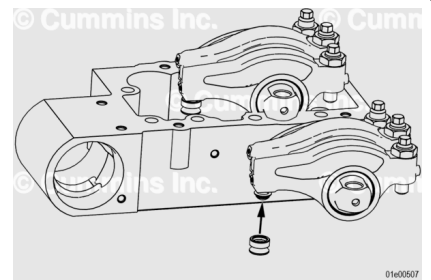
Use an old push tube socket end to support and align the ball for proper seating.

Use a press to seat the new ball into the cavity in the rocker lever.

The lip on the ball **must** be flush with the surface on the rocker lever.

⚠ CAUTION ⚠

Verify the o-ring is positioned in the groove of the socket properly. If the o-ring is installed on the ball, the socket can drop off during assembly on the engine.



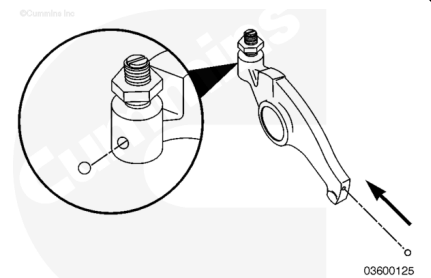
Use clean engine oil to lubricate the o-ring, socket and ball.

The o-ring is **only** used to retain the socket to the rocker lever during assembly.

Install the new socket on the ball.

If removed, install drive plugs into oil drillings in the rocker levers.

Contact a Cummins® Authorized Repair Location for the proper sealing mechanism part number.

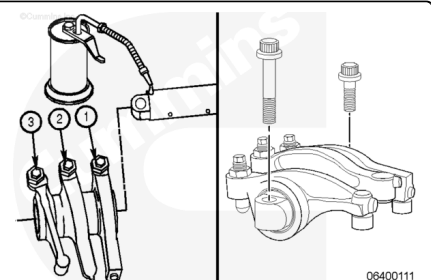


Install

with Mechanically Actuated Injector

⚠ CAUTION ⚠

The oil drillings in the shaft must be in alignment with the oil drillings in the rocker levers.



⚠ CAUTION ⚠

The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.

Use clean oil to lubricate the shaft.

Slide the rocker levers on the shaft in the sequence shown.

- Exhaust lever (1)
- Injector lever (2)
- Intake lever (3)

Lubricate the threads, rocker sockets, and crosshead pads with clean engine oil.

Install the washers and capscrews.

Verify the ring dowel that aligns the rocker shaft is installed in the housing.

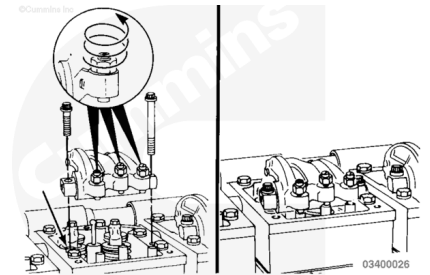
Verify the adjusting screws are loose.

Position the rocker lever assembly on the housing. Install the capscrews and finger-tighten two or three revolutions.

Align the pushrod sockets with the adjusting screws.

Tighten the capscrews.

Torque Value: 280 N·m [207 ft-lb]



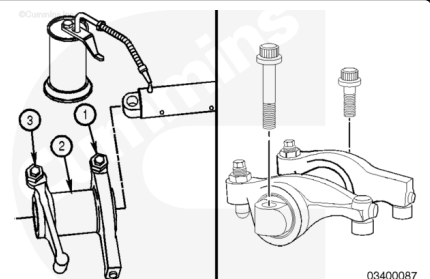
with Electronically Actuated Injector

⚠ CAUTION ⚠

The oil drillings in the shaft must be in alignment with the oil drillings in the rocker levers.

⚠ CAUTION ⚠

The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.



Use clean oil to lubricate the shaft.

Install the exhaust lever (1), spacer (2), and intake lever (3) onto the shaft. The oil drillings in the shaft **must** align with the drillings in the levers.

Note : The plug in the end of the shaft **must** be on the same end as the exhaust rocker lever.

Lubricate the threads, rocker sockets, and crosshead pads with clean engine oil.

Install the washers and capscrews.

Verify the ring dowel that aligns the rocker shaft is installed in the housing.

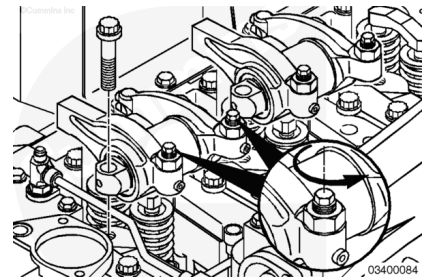
Verify the adjusting screws are loose.

Position the rocker lever assembly on the housing. Install the capscrews and finger-tighten two or three revolutions.

Align the pushrod sockets with the adjusting screws.

Tighten the capscrews.

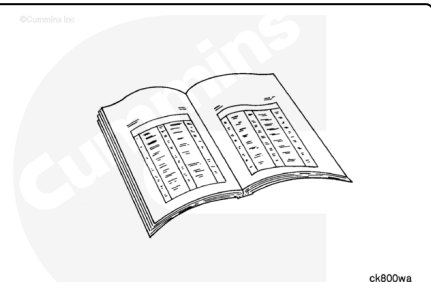
Torque Value: 280 n·m [207 ft-lb]



Finishing Steps

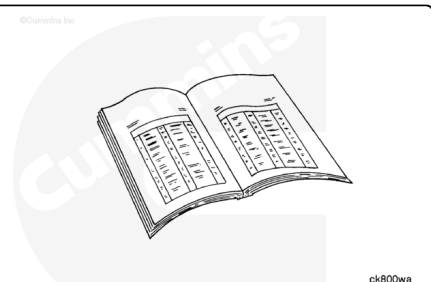
Industrial and Power Generation

- Adjust the valves and injectors. Refer to Procedure 003-006 (</qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html>).
- Install the rocker lever covers. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>).



QSK60 Marine Applications

- Adjust the valves and injectors. Refer to Procedure 003-006 (</qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html>).
- Install the rocker lever covers. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>).



- Install the aftercooler heat shields. Refer to Procedure 010-129 (/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html).

with Electronically Actuated Injector

- Adjust the overhead. Refer to Procedure 003-006 (/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html).
- Install the rocker lever covers. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html).
- Install the injector supply lines. Refer to Procedure 006-051 (/qs3/pubsys2/xml/en/procedures/56/56-006-051.html).

